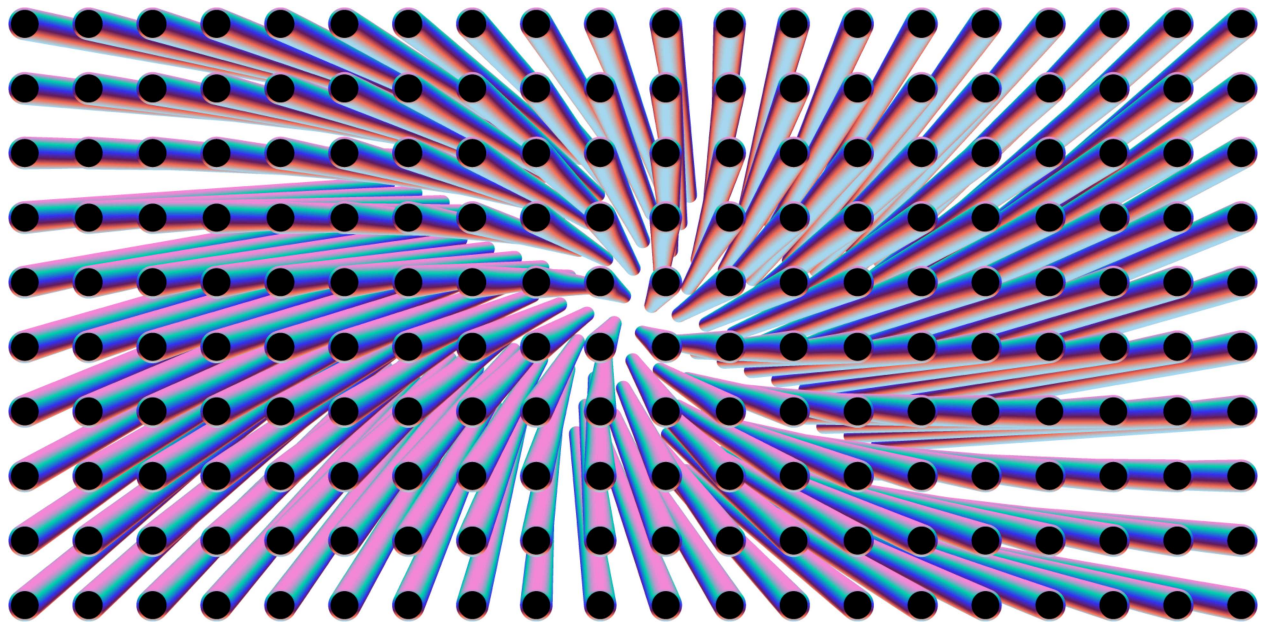


In May 2024, over 65,000 developers responded to our annual survey about coding, the technologies and tools they use and want to learn, AI, and developer experience at work. Check out the results and see what's new for Stack Overflow users.

[Developer Profile](#)[Technology](#)[AI](#)[Work](#)[Community](#)[Professional Developers](#)[Methodology](#)[Methodology.](#)

65,437

Coders from 185 countries around the world.

[Technology.](#)

62.3%

Have used JavaScript in the past year.

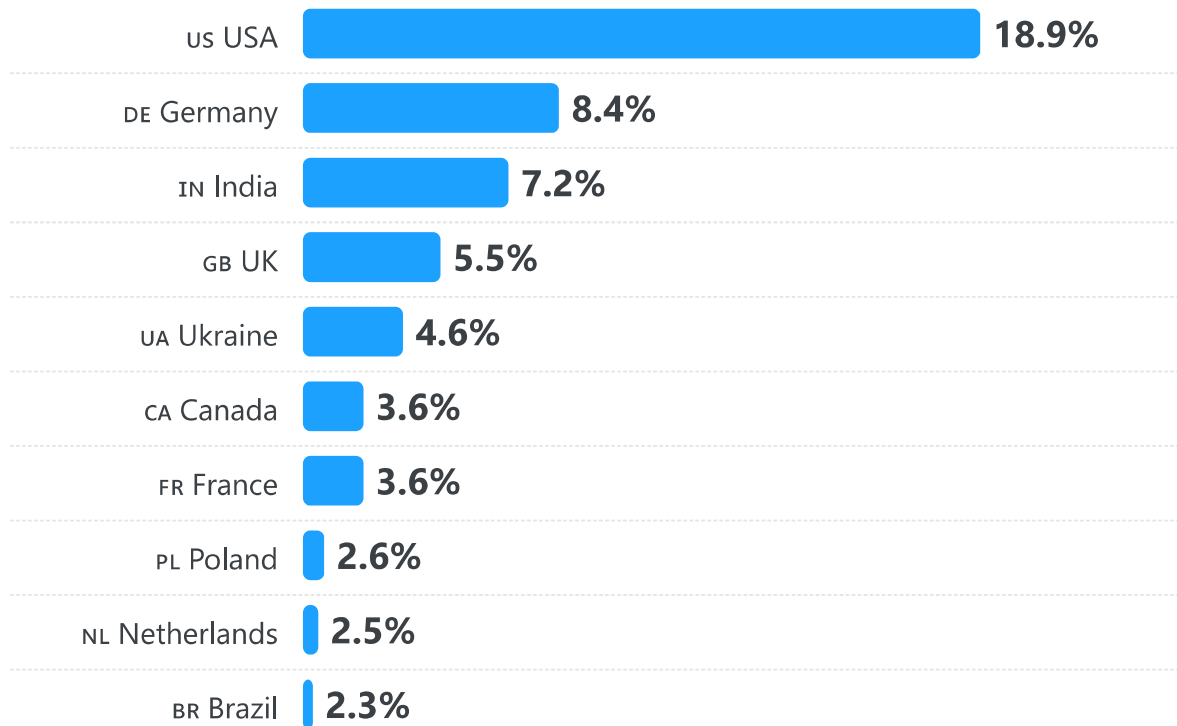
[Methodology.](#)

48,019

Respondants shared salary
information.

[Developer profile](#) → [Key territories](#)

Top 10 countries



[Community](#)

76.3%

Have a Stack Overflow account.

Advertising & Talent

Reach devs & technologists worldwide
about your product, service or
employer brand.



Now available! A suite of GenAI tools for Stack Overflow for Teams that helps connect employees to knowledge faster.

[Developer Profile](#) → [Education](#)

Educational attainment

66% of developers have a BA/BS or MA/MS degree despite only 49% of developers learning to code at school.

[Educational attainment](#) →

[Developer Profile](#) → [Learning to code](#)

Learning to code

Online resources are the top choice for developers to learn code according to 82% of responses.

Respondents 18 - 24 years old are most likely to learn code at traditional schools of all the age groups, and consistently so for the last three years.

[Learning to code](#) →

[Developer Profile](#) → [Learning to code](#)

Technical documentation preferences to learn how to code

API and SDK documents are the documentation source of choice for 90% of developers.

[Technical documentation preferences to learn how to code](#) →

[Technology](#) → [Most popular technologies](#)

Databases

PostgreSQL debuted in the developer survey in 2018 when 33% of developers reported using it, compared with the most popular option that year: MySQL, in use by 59% of developers. Six years later, PostgreSQL is used by 49% of developers and is the most popular database for the second year in a row.

[Databases](#) →

[Technology](#) → [Most popular technologies](#)

Other tools

Docker is used the most by professional developers (59%) and npm is used the most by developers learning to code (45%).

[Other tools](#) →

[Technology](#). → [Most popular technologies](#)

Asynchronous tools

Jira and Confluence top the list for most used asynchronous tools developers use for the third year.

[Asynchronous tools](#) →

[Technology](#). → [Admired and Desired](#)

Programming, scripting, and markup languages

JavaScript, Python and SQL are all highly-desired and admired programming languages, but Rust continues to be the most-admired programming language with an 83% score this year.

[Programming, scripting, and markup languages](#) →

[Technology](#). → [Admired and Desired](#)

Web frameworks and technologies

73% of developers that used it want to keep working with Svelte. Fun fact: Our team at Stack Overflow used Svelte for the first time in building our 2024 Developer Survey results site. We could go on and on about Svelte, [listen to us do just that in a interview with one of our own](#).

[Web frameworks and technologies](#) →

[Technology](#). → [Admired and Desired](#)

AI Search and Developer Tools

Most developers who use ChatGPT want to keep using it (75% Admired), while most who use Replit Ghostwrite or Tabnine are least likely to keep using it (36% Admired). If you aren't familiar with Replit, [we have an interview for you to familiarize yourself](#).

[AI Search and Developer Tools](#) →

[Technology](#). → [Worked with vs. want to work with](#)

AI Search and Developer Tools

Most developers use ChatGPT of all the AI tools, and 74% want to keep using it next year. 41% of ChatGPT users want to use GitHub Copilot next year.

[AI Search and Developer Tools](#) →

[Technology](#). → [Top paying technologies](#)

Top paying technologies

Erlang developers take the top spot this year for highest reported median salary.

[Top paying technologies](#) →

[AI](#) → [Sentiment and usage](#)

AI tools in the development process

76% of all respondents are using or are planning to use AI tools in their development process this year, an increase from last year (70%). Many more developers are currently using AI tools this year, too (62% vs. 44%).

[AI tools in the development process](#) →

[AI](#) → [Developer tools](#)

AI tools next year

In the next year, most developers agree that AI tools will be more integrated mostly in the ways they are documenting code (81%), testing code (80%), and writing code (76%).

AI tools next year →

[AI](#) → [Efficacy and Ethics](#)

Are AI tools a threat to your job

70% of professional developers do not perceive AI as a threat to their job.

Are AI tools a threat to your job →

[AI](#) → [Efficacy and Ethics](#)

Most important ethical issues for AI

Misinformation and disinformation in AI results are a top ethical concern in this year's survey for 79% of developers. Source attribution is the next top ethical concern (65%) which [Stack Overflow is working to protect for our community members](#).

Most important ethical issues for AI →

[Work](#) → [Employment](#)

Employment status

84% of respondents are working: either part-time, freelancing, or full-time.

Employment status →

[Work](#) → [Employment](#)

Work environment

The number of hybrid developers has stayed consistent this year with last year (42%), while the percentage of developers working in-person has increased (20%) for the third year in a row compared to 15% in 2022 and 16% in 2023.

Work environment →

[Work](#) → [Purchasing technology](#)

Tech endorsement factors

APIs are pretty cool, but did you know that 75% of developers are more likely to endorse your technology if you provide access to APIs? Make sure to let back-end developers and engineering managers know about the API integrations you offer: they are even more likely to endorse your tech than the average coder.

Tech endorsement factors →

[Community](#) → [Stack Overflow site use](#)

Frequency of visiting Stack Overflow

93% of respondents visit Stack Overflow at least multiple times per month if not multiple times per day.

Frequency of visiting Stack Overflow →

[Community](#) → [Participation and Feedback](#)

How you choose to use Stack Overflow

Everyone agrees that finding answers quickly is the primary reason to use Stack Overflow. Understandably, those learning to code are more likely to use Stack Overflow to learn coding skills than professional developers or other coders.

How you choose to use Stack Overflow →

[Professional Developers](#) → [Productivity Impacts](#)

Years of professional work experience

1 in 4 are 1 to 4: as in 25% of professional developers report having one to four years of work experience. The majority of respondents are early-to-mid career professionals

(nine or less years experience) and the majority of people managers are mid-to-later career professionals (10+ years experience).

Years of professional work experience →

[Professional Developers](#) → [Productivity Impacts](#)

Daily time spent searching for answers/solutions

61% of all respondents spend more than 30 minutes a day searching for answers or solutions to problems.

People managers spend less time searching than individual contributors (40% vs. 36% spend 30 minutes or less).

Daily time spent searching for answers/solutions →

Survey index

1. Developer Profile	2. Technology	3. AI	4. Work	5.
1.1. Education Educational attainment 1.2. Learning to code Learning to code Online resources to learn how to code Technical documentation preferences to learn how to code 1.3. Experience Years coding	2.1. Most popular technologies Programming, scripting, and markup languages Databases Cloud platforms Web frameworks and technologies Embedded Technologies Other frameworks and libraries Other tools	3.1. Sentiment and usage AI tools in the development process AI tool sentiment 3.2. Developer tools Benefits of AI tools Accuracy of AI tools	4.1. Employment Employment status Employment status by geography Work environment 4.2. Company info Company size 4.3. Salary	

Years coding professionally	Integrated development environment	AI tools' ability to handle complex tasks	Salary by developer type
Years of professional coding experience by developer type	Asynchronous tools	AI in the development workflow	Salary and experience by developer type
1.4. Developer roles	Synchronous tools	AI tools next year	Salary and experience by language
	Operating system		
Developer type	AI Search and Developer Tools		4.4. Purchasing technology
1.5. Key territories	2.2. Admired and Desired	3.3. Efficacy and Ethics	Influence on technology purchases
Geography	Programming, scripting, and markup languages	Are AI tools a threat to your job	Prefer to build and/or buy new tech
1.6. Demographics	Databases	Most important ethical issues for AI	Researching new tools and technologies
Age	Cloud platforms	Challenges with AI at work	Tech endorsement factors
	Web frameworks and technologies	AI at work in one year	4.5. Coding outside of work
	Embedded Technologies		Coding outside of work
	Other frameworks and libraries		
	Other tools		
	Integrated development environment		
	Asynchronous tools		
	Synchronous tools		
	AI Search and Developer Tools		
	2.3. Worked with vs. want to work with		
	Programming, scripting, and markup languages		

Databases
Cloud platforms
Web frameworks and technologies
Embedded Technologies
Other frameworks and libraries
Other tools
Integrated development environment
Asynchronous tools
Synchronous tools
AI Search and Developer Tools

2.4. Top paying technologies

Top paying technologies
Change in salaries between 2023 and 2024



Site design / logo © 2024 Stack Exchange Inc.
User contributions licensed under CC BY-SA.
Data licensed under Open Database License (ODbL).

